

Classical Observables in Gravity from Heavy-Mass Effective Theory

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Correlators and Amplitudes in the South East
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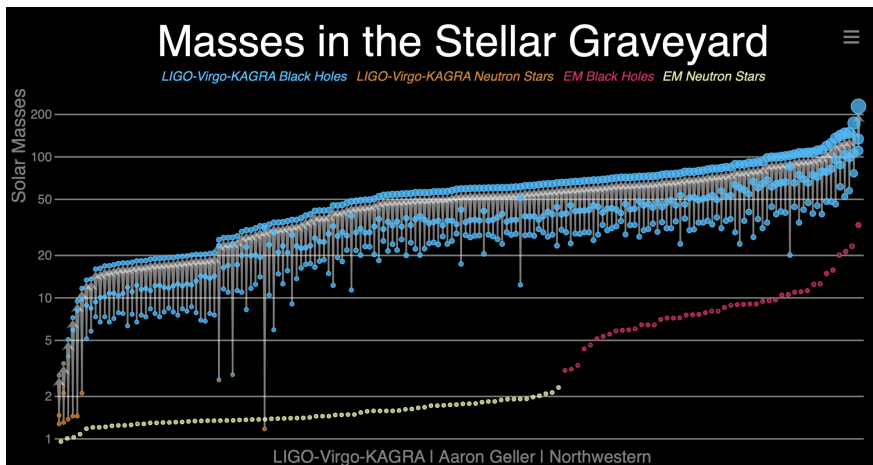
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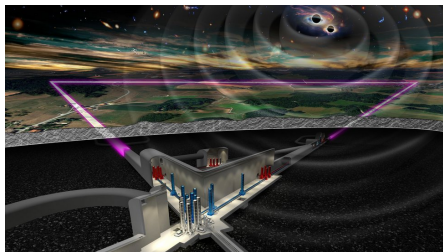
Gravitational Waves

- Experiments from LIGO-Virgo-KAGRA Collaboration
- **Binary mergers** of Black Holes and Neutron Stars
- To date, more than 200 binary mergers detected by LVK Collaboration

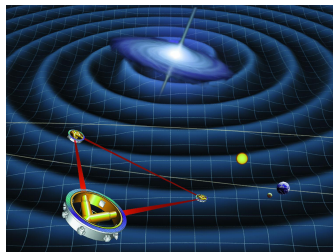


Gravitational Waves

- 3rd generation of GW observatories to start in the 2030s: Einstein Telescope, LISA, Advanced LIGO, Cosmic Explorer
- Highly increased sensitivity and larger frequency range; need for high-precision theory predictions



Einstein Telescope



LISA

From GR to EFTs

GR

Black holes are the simplest solutions: uniqueness theorems

ADM charges $(M, a, \dots)$



EFT

Point particles are uniquely characterized by few charges:

Covering of Penrose ineqs $(M, s, \dots)$

From GR to EFTs

GR

Black holes are the simplest solutions: uniqueness theorems

ADM charges $(M, \alpha, \dots)$



EFT

Point particles are uniquely characterized by few charges:

Coincident of Poincaré invariants $(M, s, \dots)$

- Black holes/neutron stars are **effectively point particles (EFT)**
- Use many **advanced amplitudes techniques** in the perturbative regime
- Post-Minkowskian (PM) expansion; weak field, all-orders in velocity
- $n\text{PM} \rightarrow G^n \rightarrow (n-1)$ loops

Classical Physics from Loops

Iwasaki; Holstein, Donoghue

- Classical physics vs quantum physics: $\hbar$
- Feynman diagram, **vertex gets $\frac{1}{\hbar}$** from expanding

$$\exp\left(\frac{i}{\hbar} \int \mathcal{L}_I(\phi_I) d^4x\right)$$

Field commutators

$$[\phi(\vec{x}), \pi(\vec{y})] = i\hbar \delta^3(\vec{x} - \vec{y})$$

- so each **propagator gets $\hbar$**

$$\langle 0|T(\phi(x)\phi(y))|0\rangle = \int_k \frac{i\hbar e^{ik(x-y)}}{k^2 - \frac{m^2}{\hbar^2} + i\varepsilon}$$

Classical Physics from Loops

- Each **vertex** V gets $\hbar^{-1}$
- Each **propagator** I gets $\hbar^{+1}$
- Diagram with V vertices and I internal lines, the number of **loops** is $L = I - V + 1$
- Diagrams get factor of $\hbar^{I-V+1} = \hbar^L$
- So, if we want classical physics, we can just stay at tree level!
- Not quite...

Classical Physics from Loops

- There is also $\hbar$ hidden in the kinematics of **massless particles**:

$$p^\mu = \hbar k^\mu$$

- So a loop contribution is not just:

$$\text{loop} \sim \hbar.$$

- It can be:

$$\text{loop} \sim \hbar \times f(\hbar, k, m),$$

- and the function $f(\hbar, k, m)$ may itself contain hidden inverse powers of $\hbar$.
- So, we need to watch out for interplay between **massive and massless particles**

Classical Physics from Loops: Example

- Example: one-loop QED calculation of the matrix element of the energy-momentum tensor

$$\langle p_2 | T_{\mu\nu}(0) | p_1 \rangle = \frac{1}{\sqrt{4E_2E_1}} \left[2P_\mu P_\nu F_1(q^2) + (q_\mu q_\nu - \eta_{\mu\nu} q^2) F_2(q^2) \right].$$

- One-loop corrections:

$$F_1(q^2) = 1 + \frac{e^2}{16\pi^2} \frac{q^2}{m^2} \left(\frac{3}{4} \frac{m\pi^2}{\sqrt{-q^2}} - \frac{8}{3} + 2 \log \frac{-q^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots$$

$$q^2 = \hbar^2 k^2, \quad e^2 \rightarrow \frac{e^2}{\hbar}.$$

Classical Physics from Loops

- One-loop **corrections**:

$$F_1(q^2) = 1 + \frac{e^2}{16\pi^2} \frac{q^2}{m^2} \left(\frac{3}{4} \frac{m\pi^2}{\sqrt{-q^2}} - \frac{8}{3} + 2 \log \frac{-q^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots$$

- Restore factors of $\hbar$:

$$q^2 = \hbar^2 k^2, \quad e^2 \rightarrow \frac{e^2}{\hbar}.$$

$$\begin{aligned} F_1(k^2) &= 1 + \frac{e^2}{16\pi^2 \hbar} \frac{\hbar^2 k^2}{m^2} \left(\frac{3}{4} \frac{m\pi^2}{\sqrt{-\hbar^2 k^2}} - \frac{8}{3} + 2 \log \frac{-\hbar^2 k^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots \\ &= 1 + \frac{3e^2 \sqrt{-k^2}}{64m} + \frac{\hbar e^2 k^2}{16\pi^2 m^2} \left(-\frac{8}{3} + 2 \log \frac{-\hbar^2 k^2}{m^2} - \frac{4}{3} \log \frac{\lambda}{m} \right) + \dots \end{aligned}$$

- Classical contribution from a loop diagram!**
- Loop-counting by itself misses the hidden $\hbar$ -dependence in **nonanalytic momentum terms**
- Massless propagators can **generate classical long-range pieces from loops**

Classical Observables from QFT: KMOC

Kosower, Maybee, O'Connell; Brown's Thesis

- Model our massive objects using localised scalar wavepackets,

$$|\psi\rangle := \int d\Phi(p) e^{ip \cdot b} \phi(p) |p\rangle \quad (1)$$

- We need to construct **two-particle states**: take the product of two different copies of the above equation:

$$|\Psi\rangle_{\text{in}} = |\psi_1\rangle \otimes |\psi_2\rangle := \int d\Phi(p_1) d\Phi(p_2) e^{i(p_1 \cdot b_1 + p_2 \cdot b_2)} \phi(p_1) \phi(p_2) |p_1, p_2\rangle \quad (2)$$

- The variance in the momentum of the wavepacket is proportional to ξ :
'spread' of wavepacket

Classical Observables from QFT

For this product to describe two localised classical particles, we require that:

- The wavepackets be **well separated compared to their spatial spread**
- The wavepackets are well localised in momentum space; that they **do not spread out too quickly with time**

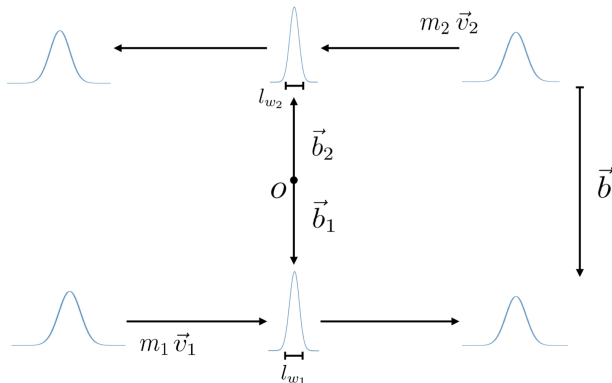
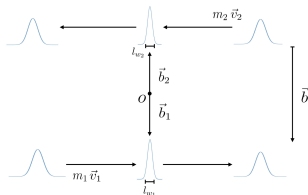


Figure: from Graham Brown's Thesis

Classical Observables from QFT



These can both be expressed as the 'Goldilocks' conditions:

$$l_{c_i} := \frac{1}{m_i} \ll l_{w_i} \ll |b| \quad (3)$$

- Quantum effects are suppressed by making the wavelength of the particle much smaller than the width of the wavepacket
- Wavepackets are well separated from each other, and we can treat the particles as point-like and independent

Classical Limit

$$l_{c_i} := \frac{1}{m_i} \ll l_{w_i} \ll |b| \quad (4)$$

- Reintroduce factors of $\hbar$:

$$\sqrt{\xi_i} = \frac{l_{c_i}}{l_{w_i}} = \frac{\hbar}{m_i l_{w_i}}, \quad (5)$$

so the **classical limit** $\xi \rightarrow 0$ can be written as either

- $\hbar \rightarrow 0$
- heavy-mass limit $m_i \rightarrow \infty$, HEFT expansion

Binary Dynamics from Amplitudes

- Given these states, we can compute classical observables
- Example: **impulse** (change in momentum)
- Incoming** momentum:

$$\langle p_{\text{in},1}^\mu \rangle = {}_{\text{in}} \langle \psi | \mathbb{P}_1^\mu | \psi \rangle_{\text{in}} \quad (6)$$

- Outgoing** momentum:

$$\langle p_{\text{out},1}^\mu \rangle = {}_{\text{out}} \langle \psi | \mathbb{P}_1^\mu | \psi \rangle_{\text{out}} \quad (7)$$

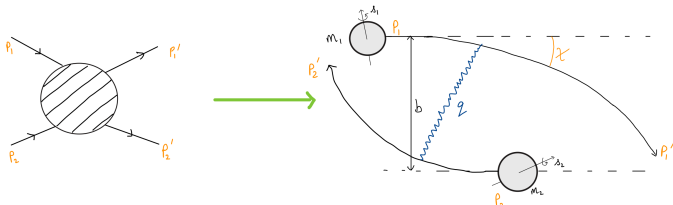
$$= {}_{\text{in}} \langle \psi | S^\dagger \mathbb{P}_1^\mu S | \psi \rangle_{\text{in}} \quad (8)$$

- Impulse:**

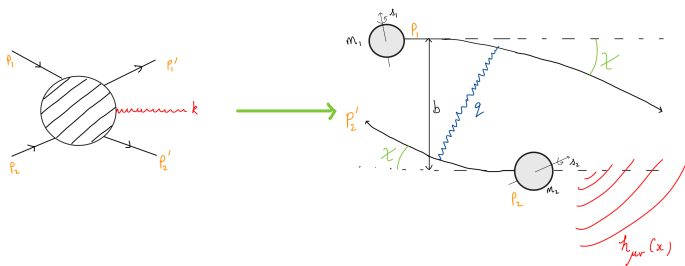
$$\langle \Delta p_1^\mu \rangle = {}_{\text{in}} \langle \psi | S^\dagger \mathbb{P}_1^\mu S - \mathbb{P}_1^\mu | \psi \rangle_{\text{in}} \quad (9)$$

Binary Dynamics from Amplitudes

- In the **conservative regime**, bending angle χ from **4-point amplitudes**



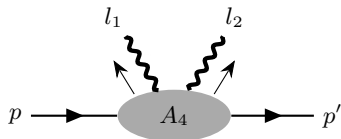
- In the **dissipative regime**, **bremstrahlung/waveforms** from **5-point amplitudes**



Gravitational Compton Scattering from HEFT

Brandhuber, Brown, Chen, De Angelis, Gowdy, Travaglini

Tree-Level Compton Scattering



which evaluates as

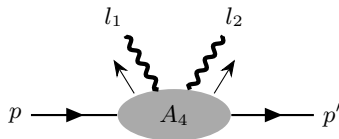
$$A_4 = \frac{i\kappa^2}{16} \left(\frac{N_{12}^2}{D_{12}} + \frac{N_{21}^2}{D_{21}} + \frac{N_{[12]}^2}{D} \right), \quad (10)$$

and can be rewritten as

$$A_4 = \frac{i\kappa^2}{16} \frac{(D_{12}N_{21} + D_{21}N_{12})^2}{D_{12}D_{12}D} = 4i\kappa^2 \frac{(p \cdot F_1 \cdot F_2 \cdot p)^2}{D_{12}D_{21}D}, \quad (11)$$

Kinematics and Propagator Expansion

Damgaard, Haddad, Helset; Brandhuber, Chen, Travaglini, Wen



- Use kinematics

$$p = \bar{p} + \frac{q}{2}, \quad p' = \bar{p} - \frac{q}{2}, \quad q = \ell_1 + \ell_2, \quad \bar{p} \cdot q = 0 \quad (12)$$

- to rewrite massive propagators

$$D_{12} = -2(p \cdot \ell_1) + i\varepsilon, \quad D_{21} = -2(p \cdot \ell_2) + i\varepsilon \quad (13)$$

- in terms of **barred variables**

$$\begin{aligned} \frac{1}{D_{12}} &= \frac{1}{-2(\bar{p} \cdot \ell_1) + i\varepsilon} + \frac{q^2}{2(-2\bar{p} \cdot \ell_1 + i\varepsilon)^2} + \dots, \\ \frac{1}{D_{21}} &= \frac{1}{2(\bar{p} \cdot \ell_1) + i\varepsilon} + \frac{q^2}{2(2\bar{p} \cdot \ell_1 + i\varepsilon)^2} + \dots, \end{aligned} \quad (14)$$

Propagator Expansion

$$A_4 = 4i\kappa^2 \frac{(p_1 \cdot F_1 \cdot F_2 \cdot p_1)^2}{D_{12} D_{21} D}, \quad (15)$$

$$\frac{1}{D_{12} D_{21} D} = -\frac{1}{(q^2 + i\varepsilon)^2} \left(\frac{1}{D_{12}} + \frac{1}{D_{21}} \right) \quad (16)$$

$$\begin{aligned} \frac{1}{D_{12}} + \frac{1}{D_{21}} &= \frac{1}{-2(\bar{p} \cdot \ell_1) + i\varepsilon} + \frac{1}{2(\bar{p} \cdot \ell_1) + i\varepsilon} \\ &+ \frac{q^2}{2(-2\bar{p} \cdot \ell_1 + i\varepsilon)^2} + \frac{q^2}{2(2\bar{p} \cdot \ell_1 + i\varepsilon)^2} + \dots, \end{aligned} \quad (17)$$

- Writing denominators as

$$\frac{1}{x \pm i\varepsilon} = \mp i\pi\delta(x) + \text{PV} \left(\frac{1}{x} \right), \quad (18)$$

then

$$\frac{1}{-x + i\varepsilon} + \frac{1}{x + i\varepsilon} = -2\pi i\delta(x) \quad (19)$$

Amplitude after Propagator Expansion

$$\frac{1}{D_{12}D_{21}D} = -\frac{1}{(q^2 + i\varepsilon)^2} \left(\frac{1}{D_{12}} + \frac{1}{D_{21}} \right) = \frac{1}{(q^2 + i\varepsilon)^2} \left[i\pi \delta(\bar{p} \cdot \ell_1) - \frac{q^2}{4(\bar{p} \cdot \ell_1)^2} \right] + \dots \quad (20)$$

to see that A_4 splits into two terms

The diagram shows the expansion of the amplitude A_4 . On the left, a grey circle labeled A_4 has an incoming line from the left labeled p and an outgoing line to the right labeled p' . Two wavy lines, labeled ℓ_1 and ℓ_2 , emerge from the top of the circle. This is set equal to the sum of two terms. The first term consists of two grey circles, each labeled H , connected by a vertical red line. The left H has an incoming line p and an outgoing wavy line ℓ_1 pointing upwards. The right H has an incoming wavy line ℓ_2 pointing upwards and an outgoing line p' . This term is labeled "Hyperclassical $\mathcal{O}(\bar{m}^3)$ ". The second term is a single grey circle labeled H with an incoming line p and an outgoing line p' , and two wavy lines ℓ_1 and ℓ_2 emerging from the top. This term is labeled "Classical $\mathcal{O}(\bar{m}^2)$ ". The entire equation is labeled (21).

- Hyperclassical contributions $\rightarrow \infty$ in the classical limit $\bar{m} \rightarrow \infty$; unphysical
- Amplitudes have hyperclassical contributions, but cancel in anything observable (e.g. impulse)

Conclusion & Outlook

- **Massless** propagators can generate **classical** long-range physics from **loop diagrams**
- Classical observables can be found from QFT in the **classical limit**: $\hbar \rightarrow 0$ or $m \rightarrow \infty$
- **HEFT** distinguishes hyperclassical, classical, quantum contributions at an earlier stage
- Future work will focus on different theories, different diagrams, to better understand the **Magnus expansion** (see Pablo's talk)

Thank you!